

Name: Anthony Claudet

Date: 5/3/25

Class: Programming for Engineers

Element B: Documentation and Analysis of Current and Prior Design Attempts

Current or Prior Design Attempt #1 - Energy Conservation Scheduling

Source

Based on general recommendations from [Green City Times](#), which discuss basic energy-saving behaviors like manually turning off appliances and using programmable thermostats.

Image



Image source: Green City Times, Sustainable Smart Home Systems.

Details

- Homeowners manually set schedules for device use (e.g., turning off heating or lights at specific times).
- Requires user planning and strict adherence to schedules.
- No automatic detection of peak hours or dynamic energy pricing changes.

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Analysis

While manual scheduling helped reduce some unnecessary energy consumption, it heavily relied on user behavior, which was inconsistent. Most users forgot to follow schedules or override them for convenience, leading to continued energy waste and missed optimization opportunities. The design requirement of automatic responsiveness was not met.

Prior Design Attempt #2 - Static smart plugs and timers

Source

This was discussed in [ResearchGate's study on Smart Homes and Energy Conservation](#).

Image

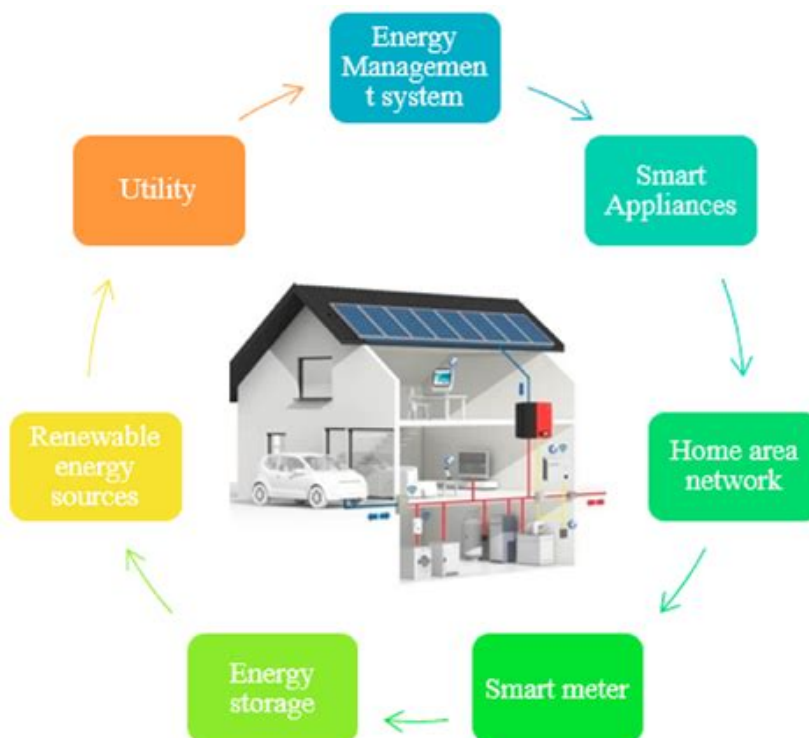


Image source: ResearchGate, Impact of Smart Homes on Energy Conservation.

Details

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- Smart plugs and timers automatically shut off devices at preset times.
- Devices can be turned on/off remotely but generally don't adjust automatically based on peak energy hours or current load.

Analysis

Static smart plugs helped reduce "phantom" energy usage by powering off devices not in use, but timing was rigid (ResearchGate). Devices didn't adjust based on real-time energy prices, peak hours, or household activity. The system lacked adaptability, a key design requirement for true smart energy management.

Prior Design Attempt #3 - Full smart home systems without AI optimization

Source

Based on modern smart home energy management systems overviewed in [ScienceDirect's article](#).

Image

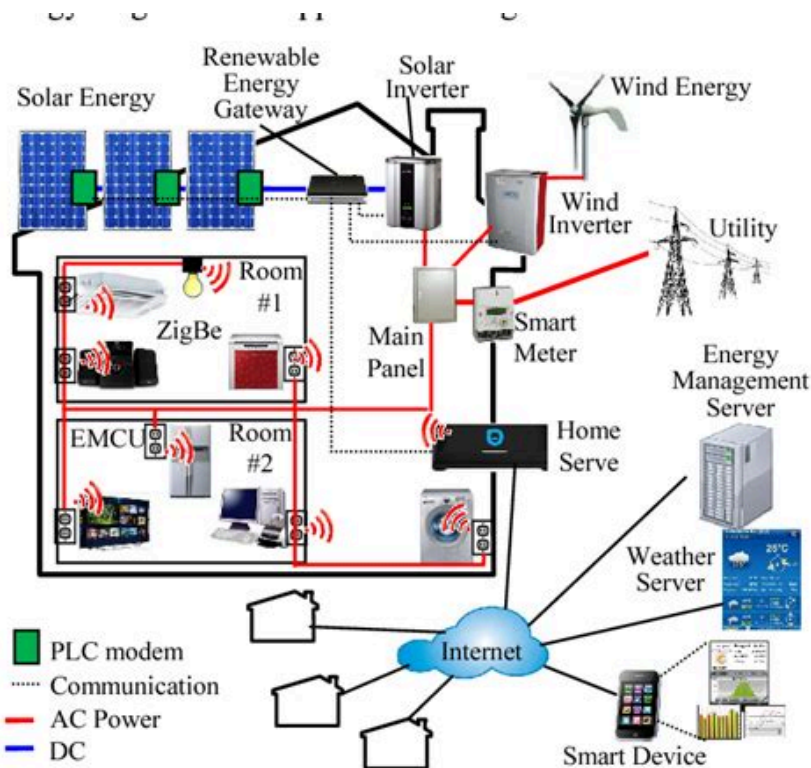


Fig. 1. Architecture of smart home energy management (HEMS)

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Image source: ScienceDirect, Home Energy Management Systems.

Details

- **Smart home hubs (Google Nest, Amazon Echo, etc) connect devices.**
- **Devices are linked but operate on fixed routines or basic rules, not real-time energy optimization.**
- **Limited ability to prioritize devices dynamically based on essentiality or energy loads.**

Analysis

Even modern smart homes lacked true intelligent energy optimization without specialized AI or adaptive learning algorithms (ScienceDirect). Systems operated devices based on preconfigured rules rather than analyzing current conditions (peak demand, occupant presence). Therefore, they failed to meet the design requirement for dynamic, efficient energy balancing.

